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Original research article

Carcass size and ground substrate drive detection rates of avian carcasses by human surveyors and a dog team

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ABSTRACT

Accurate avian mortality estimates are essential for understanding anthropogenic impacts to bird populations and informing conservation strategies. Carcass surveys are commonly conducted by human surveyors or by detection dogs, but the factors influencing surveyor detection abilities have not been fully explored. In this study, we conducted two years of detection trials in the semi-arid high desert of southern New Mexico, USA, testing 27 human surveyors and one conservation detection dog across 1096 trials with 238 carcasses representing 50 avian species. We directly compared detection abilities between surveyor types (human and dog) and identified key factors influencing detection probabilities. The conservation detection dog exhibited a significantly higher detection probability (mean = 0.87) than human surveyors (mean = 0.49, individuals ranged 0.25–0.71), consistent with previous studies. Detection probabilities for both surveyor types were influenced by carcass size and ground substrate; detection probability was higher for larger carcasses and areas with lower vegetative complexity. We discuss our results in the context of common tradeoffs faced by managers in designing carcass surveys and how guidance may vary under different scenarios. Broadly, our study provides valuable insight that can enhance wildlife mortality monitoring, ensuring more accurate mortality estimates to inform management and conservation efforts.

1. Introduction

Anthropogenic activities are altering landscapes and causing unprecedented avian declines (Loss et al., 2015; Rosenberg et al., 2019). Collisions with artificial structures represent a major threat to birds—in the United States, an estimated 1 billion birds die annually from building collisions (Kornreich et al., 2024; Loss et al., 2014a), 12–64 million from power line collisions and electrocutions (Loss et al., 2014b), and 573 thousand from collisions with wind turbines (Smallwood, 2013). These estimates likely underestimate total mortality due to recent infrastructure expansion and a lack of systematic monitoring (Loss et al., 2015; Potter, 2023). Accurately estimating avian mortality is vital for understanding the impact of artificial structures on avian mortality and developing

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evidence-based solutions that mitigate avian mortality and guide conservation actions (Huso, 2019).

Carcass surveys are a common method used to estimate avian mortality associated with chronic (e.g., buildings, energy infrastructure collisions; Borner et al., 2017; Riding and Loss, 2018) and acute mortality events attributed to disease or poor migration conditions (Avery-Gomm et al., 2024; Jones et al., 2025). Increased awareness of anthropogenic impacts on avian populations, combined with emerging novel threats such as climate change and zoonotic diseases (Prosser et al., 2023), has highlighted the need to standardize and more accurately quantify associated mortality (take) (Conkling et al., 2022). While standardized protocols exist for monitoring live bird populations, such as the North American Breeding Bird Survey (Sauer et al., 2017) and the North American Marsh Bird Monitoring Protocol (Conway, 2011), carcass surveys have historically lacked similar standardization (Huso, 2019). Like population surveys in live birds, carcass surveys are subject to multiple sources of bias that can affect abundance or density estimates. These include: imperfect detection, where not all carcasses are found by surveyors (also known as "searcher efficiency"); scavenger removal, where carcasses are removed before surveys occur; and crippling bias, where injured birds move from inside to outside the survey area (Bernardino et al., 2013; Borner et al., 2017; Riding and Loss, 2018). The impact of these biases depends on factors such as surveyor, survey site structure and conditions, scavenger community, and time of year (Barrientos et al., 2018; Riding and Loss, 2018). Developing carcass survey designs that can correct or minimize sources of bias and enable comparisons across studies is essential for producing reliable mortality estimates to better inform conservation efforts.

Detection trials, in which a known number of carcasses are placed at survey sites prior to surveys, are used to correct for surveyor detection in mortality estimates. Traditionally, carcass surveys have been conducted by human surveyors, but the use of trained detection dogs has increased in recent years (Mathews et al., 2013; Nilsson et al., 2023). With highly sensitive olfactory systems, dogs can detect specific scents at extremely low concentrations (MacKay et al., 2008). In previous studies, dogs detected 92% of house sparrow (*Passer domesticus*) carcasses versus 45% found by humans (Homan et al., 2001), 96% of common quail (*Coturnix coturnix*) carcasses versus 9% found by humans (Paula et al., 2011), and 69% of small carcasses ($\leq$100 g) versus 40% found by humans (Reyes et al., 2016). Though these studies suggest that dogs have higher detection probabilities than humans, few studies have directly evaluated factors that may influence these differences (Domínguez del Valle et al., 2020).

Broadly, factors related to survey design, site conditions, and weather are thought to influence carcass detection probability, though evidence remains mixed for both human and dog-based surveys. For human surveyors, most studies indicate that detection improves with experience (Linz et al., 2006; Ponce et al., 2010; Riding and Loss, 2018) and carcass size (Barrientos et al., 2018; Morrison, 2002), and declines with vegetation density (Aschwanden et al., 2018; Morrison, 2002). In contrast, evidence for carcass size and vegetation cover influencing detection by dogs is mixed (Domínguez del Valle et al., 2020; Mathews et al., 2013; Nilsson et al., 2023). Weather conditions like wind and temperature can affect scent dispersal and odor intensity, potentially influencing detection by dogs, though findings are based on limited trials under narrow conditions (Homan et al., 2001; Mathews et al., 2013; Reyes et al., 2016). Thus, comparative detection trials in which the true number of carcasses is known and multiple surveyor types are exposed to the same conditions are a powerful tool to elucidate influences on detection probability.



Fig. 1. Avian carcass survey sites (white dots) used in detection trials on White Sands Missile Range (WSMR) in southern New Mexico, USA, fall 2022 and fall 2023 (left). The photos at right show examples of the variety of ground cover on survey sites including pavement (top), mowed lawns (middle), and native desert scrub (bottom). Photos by Kelley C. Boland.

In this study, we conducted detection trials of avian carcasses using human surveyors and a detection dog team. Our objectives were to: (1) estimate detection probabilities of carcasses by both human and dog surveyors; and (2) assess the influence of variables related to olfaction, vision, and effort on carcass detection to improve mortality estimates, survey designs, and management decisions. Specifically, we tested the following predictions based on previous research: (i) dogs have higher detection probabilities than humans; (ii) human detection is most strongly influenced by visual cues (such as carcass size and ground substrate), fatigue, and experience; (iii) dog detection is most strongly influenced by factors affecting odor production and scenting ability, including wind speed, carcass condition, and temperature. We predicted human detection probability would increase as carcass size increased and throughout the survey season, and human detection would decrease as vegetative complexity and time of day increased, whereas dog detection would remain high and unaffected by these variables. Finally, we predicted dog detection probability would be highest in light wind speeds and low temperatures, while human detection would remain relatively low and unaffected by these environmental conditions.

2. Materials and methods

2.1. Study area

We conducted our study on White Sands Missile Range (WSMR), in the Tularosa Basin of south-central New Mexico (32.387531° N, 106.479650° W). The climate is characterized as semi-arid high desert, with a mean annual rainfall of 25 cm; during fall (August–November), temperatures generally range from 12 to 30 °C, with mean temperatures around 20 °C (data from WSMR Meteorological Branch). Ground cover on survey sites varied from highly modified surfaces (e.g., xeriscaping, paved parking lots, manicured grassy parks) to areas with minimal or no modification including irregularly mowed weedy sites dominated by goathead (*Tribulus terrestris*), as well as native desert scrub characterized by honey mesquite (*Prosopis glandulosa*), soaptree yucca (*Yucca elata*) and fourwing saltbush (*Atriplex canescens*) (Fig. 1).

2.2. Detection trial methods

To evaluate variation in detection probability by surveyor type, human and dog, we conducted blind detection trials by placing avian carcasses at survey sites without surveyor knowledge. For each surveyor, we defined detection probability as the number of carcasses detected out of the total number of carcasses available, while a trial represented an instance in which a surveyor had the opportunity to detect a carcass.

From 1 September to 25 October in 2022 and 2023, aligning with peak passerine migration in our region (Osterhaus et al., 2025a, 2025b), we conducted blind detection trials at 25 and 23 sites, respectively (Fig. 1). Each site consisted of a 20 m-wide transect with lengths varying between sites, resulting in areas ranging from 540 to 10,565 m^2, with an average of 4065 ± 484 m^2 (SE). Surveys were performed by a human team and/or a professional conservation detection dog team; survey methods were adapted from methods used to survey avian mortality (Aschwanden et al., 2018; Borner et al., 2017; Grodsky et al., 2013). We conducted surveys between dawn and 10:00 and did not survey in rain or high wind (>25 km hr^{-1}). When logistically feasible, we performed replicate surveys, where at least two survey teams independently surveyed the same sites each morning. Each team typically covered 4–5 sites per day. Before each survey, surveyors recorded temperature (°C, ± 0.1), mean wind speed at chest height over a 30 s period (km hr^{-1}, ± 0.1), and humidity (%, ± 0.1). The blind detection trials were a part of broader effort to quantify background avian mortality during fall migration at the study area (Boland, 2024). Surveyors were aware detection trials would occur but were blind to timing and location.

2.2.1. Human surveyors

Undergraduate students at New Mexico State University (NMSU) were hired each fall and trained consistent with the methods outlined in Hager and Cosentino (2014). Prior to beginning surveys, students attended a training presentation and received a handbook detailing the project's purpose, methodology, and surveyor role. Sample photographs of avian carcasses on various ground cover types were provided to help surveyors develop a mental search image. Students participated in at least five practice surveys on pre-set routes around the NMSU campus on different ground cover types (pavement, xeriscape, and lawn), and immediate carcass detection feedback

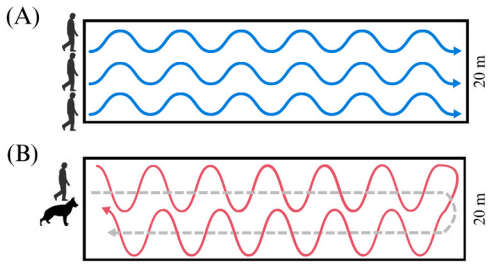


Fig. 2. Avian carcass survey method used by human and dog surveyors in southern New Mexico, USA, fall 2022 and 2023. (A) Humans walked in teams of three, zig-zagging to visually search their respective third of the survey width and not overlapping with their teammates. (B) The dog handler (dashed gray line) walked down the length of the site and back while the dog (solid red line) ranged unleashed on either side of the handler.

was given after each. In the second year of the study, returning students were paired with new participants during their first independent survey day.

Each human surveyor team was comprised of three students. At each survey site, the 20 m-wide transect was divided into three equal sections, with each surveyor assigned to a section. Walking at a moderate pace (approximately 0.8 m s^{-1}), surveyors zig-zagged in parallel within their designated sections while visually scanning for carcasses (Fig. 2A). Surveyors carried handheld GPS units (Garmin eTrex 22x, Olathe, KS) to track their individual survey paths and recorded the location of detected carcasses but left them in place; all carcasses were collected once daily surveys were complete.

2.2.2. Dog surveyors

In addition to the human surveyors, we contracted a professional conservation detection dog team to conduct carcass surveys (Find It Detection Dogs, LLC, Buena Vista, CO, USA). The dog team included one professional handler and one trained dog. The dog had four years of experience in wildlife detection studies and had been trained on a variety of scents consistent with methods outlined in Wasser et al. (2004). We used a single dog to allow for a sufficient number of trials to robustly evaluate detection performance while accommodating logistical constraints. Each year of this study, the dog was trained onsite with avian carcasses for two days prior to surveys to acclimatize to the local environment. The dog team surveyed for four weeks each season.

For each survey, the handler walked at a moderate pace while the GPS-collared (Garmin Astro, Olathe, KS) dog searched freely. The handler consulted the handheld GPS after two passes to ensure full coverage of the area, and added 1–2 additional passes to cover areas that the dog may have missed. Their search pattern followed roughly parallel transects; the dog typically ranged within 10 m of the handler (Fig. 2B). When the dog alerted to a carcass, the handler recorded its location but left it in place; carcasses were collected after daily surveys were complete. For dog safety, the dog team halted surveys if the dog exhibited heat-related stress behaviors, like excessive panting and laying in shade.

2.2.3. Detection trial design details

Sites and carcass placement locations were selected to ensure a balanced study design with both surveyor types exposed to variable carcass densities (zero to five per survey) and cover types throughout the survey season. We placed carcasses on survey sites about two hours before surveys began each morning. To minimize human scent, we handled carcasses through plastic bags and never left a dead-end scent trail at a carcass. When we placed a carcass in the field, we took a picture and recorded GPS location, carcass length (crown to tip of tail, $\pm$ 0.5 cm), carcass width (widest part of body, $\pm$ 0.5 cm), and ground substrate ("bare," bare ground or pavement; "lawn," landscaped green lawn; "short weedy," vegetation $<$15 cm height; "tall," vegetation $\geq$15 cm height). If carcass length or width could not be measured, we used the mean value for that species. To compare surveyor detection probabilities with other studies that used size classifications, we also grouped carcasses by average mass of adult birds (Wilman et al., 2014) based on size classifications used by Nilsson et al. (2023) (small, $<$24 g; medium, 24–60 g; large, $>$60 g).

We obtained carcasses that reflected a diversity of species, resulting in a range of body sizes, from calliope hummingbird (*Selasphorus calliope*) to rock dove (*Columba livia*), that were representative of resident species and fall migrants in the region. We sourced whole avian carcasses from laboratory collections and opportunistically collected fatalities (USFWS Scientific Collecting Permit MBPER0009522). We also harvested and used non-native house sparrows and European starlings (*Sturnus vulgaris*), euthanized without chemicals that could potentially emit a scent and influence the dog, pursuant to methods approved by the NMSU Institutional Animal Care and Use Committee (permit #2022–028). We stored carcasses in a -20 °C freezer and thawed carcasses overnight before use in the field. Each carcass was used in no more than two days of detection trials.

Since survey sites did not have physical markers to ensure human surveyors remained in their assigned search area, on multiple occasions a surveyor strayed from their route and detected a trial carcass placed within the path of an adjacent observer, thereby generating a "flagged" observation. To evaluate the robustness of our findings to deviations in the survey protocols, we generated three human-surveyor datasets: (1) an "inclusive" dataset that treated flagged observations as normal detections; (2) a "moderate" dataset that excluded flagged observations; whereas the (3) "conservative" dataset excluded all observations from a survey transect that included one or more flagged observations. Conversely, survey route deviations also resulted in "misses" when a surveyor failed to fully survey their assigned area and failed to detect all assigned carcasses; we treated these misses as normal non-detections in all the above datasets. We excluded trials where surveyors did not detect the carcass and the carcass could not be collected (e.g., blocked road, scavenging event, destruction by lawn mower; n = 19) and excluded any post-hoc carcass detections that occurred after a survey was completed (n = 19).

2.3. Data analysis

To examine the drivers of detection probability by human and dog surveyors, we used generalized linear mixed models with a Bernoulli distribution and a logit link function. We treated the outcome of each trial as the response variable (1 = successful detection of a trial carcass, 0 = no detection). We created two initial model sets, one for each surveyor type ("human" and "dog"), to estimate surveyor-specific detection probabilities and allow for flexibility to include covariates that were only relevant to one surveyor type.

In each model set (human and dog), we used a two-step approach to identify competitive models. First, we created a series of univariate models for each covariate and multivariate models that reflected *a priori* hypothesized relationships based on previous literature. These models were ranked using Akaike's Information Criterion adjusted for small sample size (AIC_c; Burnham and Anderson, 2002). Second, we created multivariate models by combining covariates that were represented in the best-supported models ($\Delta AIC_c \leq 2$) and were deemed informative based on the beta coefficient's 85% confidence intervals (i.e., not overlapping 0) (Arnold,

2010; Burnham and Anderson, 2002; Lawson et al., 2020). The new multivariate models explored additive and interactive (i.e., multivariate) effects that were not already represented in the *a priori* models. We then model averaged the best-supported model and all models with $\Delta AIC_c \leq 2$ using the 'model.avg' function in the *MuMIn* R package (Barton, 2023).

To develop our *a priori* hypotheses, we assigned covariates hypothesized to influence detection probability to one of three categories: olfactory, visual, and effort (Table S1). The olfactory category included wind speed (km hr^{-1}), temperature (°C), humidity (%), carcass use (first or second time used in the field), and carcass condition (whether the carcass was harvested via euthanasia or collected deceased). The visual category included carcass size (cm^2) and ground substrate (one of four categories). The effort category included survey speed (m^2 s^{-1}), survey start time (minutes since dawn), day of season (first day of surveys = day 1), year, and handler (for the dog team). We tested select variables in both human and dog model sets, as some variables aligned with multiple categories—for example, temperature could influence both scent dispersal (olfactory) and surveyor fatigue (effort). We also assessed quadratic relationships for wind speed, temperature, and day; any quadratic terms appearing in a model were accompanied by their lower-order constituent effects as additive terms. All continuous covariates were *z*-standardized across years (mean = 0.0, SD = 1.0). We assessed collinearity between covariates and variables with Pearson correlation $|r| > 0.65$ were not included within the same model. Temperature and day were correlated in the human detection data ($|r| = 0.68$); thus, we detrended the variables using linear regression (temperature ~ day), and used residual temperature variation accounting for day in all model sets.

To account for repeated sampling in our study design, we included Site ID as a random effect in all models. In the exploratory modeling phase, we also included observer as a random effect in the human-only model set but removed it from the analysis after encountering substantial convergence issues. Each model set included null and global models. We built all models using the 'glmer' function in the *lme4* package (Bates et al., 2015) in R Statistical Software (v4.2.2; R R Core Team, 2022).

To directly compare detection between human and dog surveyors, we pooled trial data by both surveyor types and created a third, post-hoc global model ("combined") using covariates deemed informative in the top models for human surveyors and the dog team. For descriptive purposes, we also calculated mean detection probabilities for individual surveyors outside of the modeling framework, and compared detection probabilities of human surveyors who participated in both years of the study using a Wilcoxon signed-rank test.

3. Results

We conducted 1096 detection trials during 236 carcass surveys. In these trials, 238 carcasses were used, representing 50 avian species (Table S2). Detection probability for human surveyors varied across the three datasets that differed in their handling of "flagged" observations from unintended human observers (n = 115). Detection probabilities were highest in the inclusive dataset (mean: 0.57; 85% CI = 0.54, 0.60; n = 721 trials), followed by the moderate dataset (0.49; 85% CI = 0.46, 0.52; n = 608), and lowest in the conservative dataset (0.47; 85% CI = 0.47, 0.51; n = 477). Ultimately, model results did not differ substantially between these datasets (Table S3, Figure S1), and results from the analysis of the moderate dataset are reported hereafter.

We tested 27 human surveyors, and individual detection probabilities varied widely from 0.25 to 0.71 (Fig. 3). Of the five observers that returned to survey in the second year, only one showed a significant ($\alpha = 0.05$) increase in detection from 2022 to 2023 ($p < 0.01$), while the other four showed no significant change between years. Detection probability for the dog team (0.86; 85% CI = 0.84, 0.89; n = 331) was higher than that of humans.

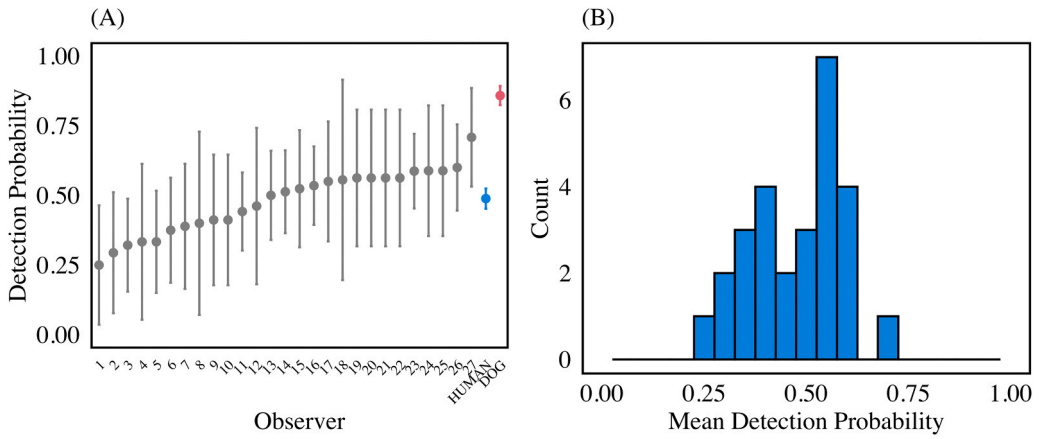


Fig. 3. Detection probabilities of avian carcasses by human surveyors in southern New Mexico, USA, fall 2022 and 2023 (n = 27). (A) Mean detection probability for each human observer, calculated as the proportion of trial carcasses detected out of the total placed, is shown with 85% confidence intervals; overall mean detection probability for human and dog surveyors are indicated in blue and red, respectively. (B) Distribution of mean detection probabilities across all human observers.

3.1. Drivers of detection by humans

The model selection results from 26 models in the first step yielded two competitive models (Table 1), and did not indicate support for novel covariate combinations not represented in our *a priori* hypotheses. Model-averaged coefficient estimates from the two competitive models indicated human detection was positively associated with carcass size and day of season, and negatively associated with substrate, wind speed, survey speed, and survey start time (Table S4, Fig. 4A). Carcass size and substrate had the strongest effects on human detection probability; detection increased with carcass size and was similar on bare ground and lawn but decreased on short weedy and tall substrates (Fig. 4A). When grouped into size classes, mean human detection probability was 0.30 (85% CI = 0.25, 0.35; n = 178) for small carcasses, 0.47 (85% CI = 0.42, 0.52; n = 228) for medium carcasses, and 0.67 (85% CI = 0.62, 0.72; n = 202) for large carcasses. Although not statistically significant, human detection probability appeared to increase over the 2022 season, while it started higher and remained relatively stable throughout the 2023 season (Fig. 5).

3.2. Drivers of detection by dog team

In the first model building step, competitive models indicated support for effects of wind, substrate, carcass size, and survey speed, so we introduced two new models to the final set: one novel additive model and one including a substrate × carcass size interaction. Ultimately, 33 models were fit to the dog detection data and one competitive model emerged (w_i: 0.77). The top-ranked model included the covariates wind speed (linear and quadratic), carcass size, substrate, and survey speed (Table 1). Based on 85% confidence intervals, all parameters except wind speed were significant (Table S4); coefficient estimates indicated dog detection was positively associated with carcass size and negatively associated with non-bare ground substrates and survey speed (Fig. 4B). Dog detection probability followed a quadratic relationship with wind speed, although this appeared to be driven by two surveys with relatively strong winds. Carcass size and substrate had the strongest effects on dog detection probability; detection increased with carcass size and was similar on bare ground and lawn but decreased on short weedy and tall substrates (Fig. 4B). When grouped into size classes, mean dog detection probability was 0.85 (85% CI = 0.79, 0.90; n = 96) for small carcasses, 0.82 (85% CI = 0.76, 0.87; n = 119) for medium carcasses, and 0.91 (85% CI = 0.86, 0.94; n = 116) for large carcasses.

3.3. Detection by humans and dog team, combined

To directly compare detection between human and dog surveyors, we fit a global model that included surveyor type (human or dog) and its interactions with the variables deemed informative in the top models for humans and the dog team – carcass size, substrate, wind speed (linear and quadratic), survey speed, start time, and day of season. Predicted detection probabilities for each surveyor type, presented side by side in Fig. 6, provide a tool for managers to evaluate relative survey performance across conditions.

4. Discussion

In this study, we compared the ability of human and dog surveyors to detect avian carcasses over two years of detection trials. As expected, the dog team showed higher detection probability; unexpectedly, both human and dog surveyors were most strongly influenced by the same variables: carcass size and ground substrate. Our study builds upon previous research by directly comparing surveyor types and addressing limitations of earlier work that lacked blind trials and site-specific weather data (Domínguez del Valle et al., 2020). Our findings allow for improved correction factors to account for imperfect detection in avian mortality estimates and provide insights for optimizing survey protocols when using human or dog surveyors, ultimately supporting more reliable mortality estimates to guide management and conservation decisions.

Detection probability estimates for human surveyors in our study were on the higher end of the range reported in the literature (typically 0.14–0.54; e.g., Arnett, 2006; Ponce et al., 2010; Smallwood et al., 2020). Relatively high detection probabilities by our human surveyors may reflect the lack of tall vegetation at most of our survey sites—high detection probabilities of human surveyors (0.70–0.80) have been reported in studies with small search areas or low vegetation cover (Aschwanden et al., 2018; Linz et al., 2006; Riding and Loss, 2018). Though mean human surveyor detection probability was relatively high, variation among human surveyors mirrored that observed by Ponce et al. (2010), indicating some individuals demonstrate strong detection abilities.

Detection probability estimates for the dog team in our study were also within the range reported in existing studies and were

Table 1
Competitive models and the first non-competitive model for each model set (human and dog) for avian carcass detection trials in southern New Mexico, USA, fall 2022 and 2023. For each model, we specify the number of parameters (K), the difference in Akaike's Information Criterion corrected for small sample size with the top model (ΔAIC_c), AIC_c model weight (w_i), and model deviance (DEV). All models included a random effect of Site ID.

Model Set	Fixed Effects	K	ΔAIC_c	w_i	DEV
Human	Wind_Speed + Temp + Substrate + Size + Speed + Start_Time + Day + Year + Day × Year	13	0.000	0.544	683.2
	Wind_Speed + Temp + Substrate + Size + Survey_Speed + Start_Time	10	1.047	0.322	690.5
	Substrate + Size + Survey_Speed + Start_Time + Day + Year	10	3.385	0.100	692.8
Dog	Wind_Speed + $Wind_Speed^2$ + Size + Substrate + Survey_Speed	9	0.000	0.773	229.1
	Wind_Speed + $Wind_Speed^2$ + Size + Substrate + Survey_Speed + Size × Substrate	12	3.616	0.127	227.0

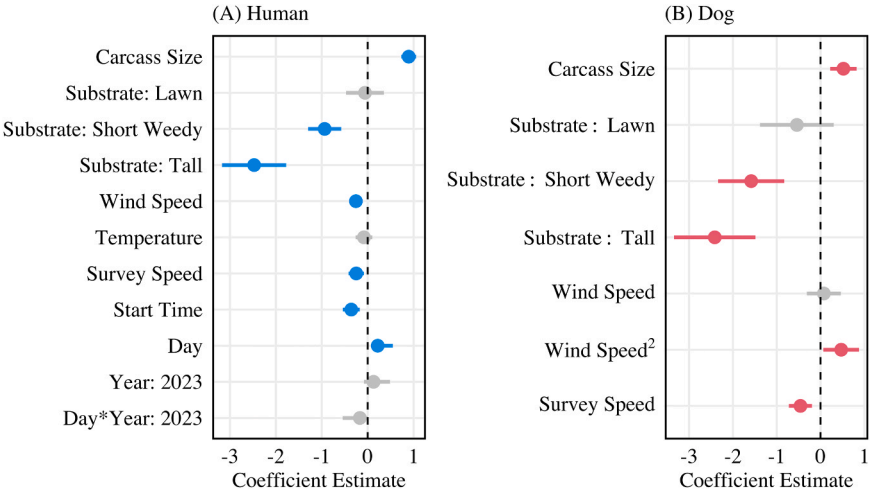


Fig. 4. Factors influencing detection probability of avian carcasses in southern New Mexico, USA, fall 2022 and 2023. Model coefficient estimates with 85% confidence intervals are shown for (A) human surveyors (model-averaged from the two competitive models) and (B) conservation detection dog surveyor (single competitive model). Significant variables are highlighted in color, with non-significant variables in gray. Categorical variables are relative to the reference level (Substrate = Bare; Year = 2022).

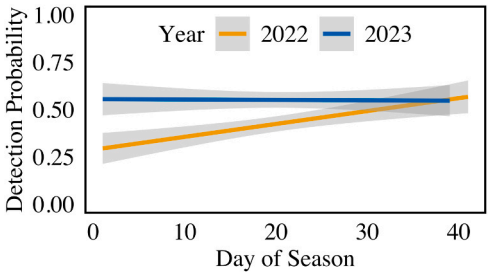


Fig. 5. Detection probability of avian carcasses by human surveyors in southern New Mexico, USA, in each fall season, 2022 and 2023. Gray shaded areas represent 85% confidence intervals. In 2023, human surveyors received improved pre-season training.

consistent with values reviewed by Barrientos et al. (2018). While evaluating a single dog limits generalization, the consistency of our estimate with prior work suggests our findings align with broader patterns reported in the literature. However, dog detection performance is known to vary based on training, experience, and handler skill (Caldicott et al., 2024; Chaber et al., 2022; Reindl-Thompson et al., 2006). This highlights the importance of selecting well-trained dogs and experienced handlers when designing surveys.

Our findings indicate that human detection of avian carcasses is most strongly influenced by visual cues, particularly carcass size and ground substrate, as well as by conditions likely to affect observer fatigue. Despite all trial carcasses in our study being relatively small bodied (<28 cm length), larger carcasses were detected more frequently than smaller carcasses, consistent with previous research (Borner et al., 2017; Ponce et al., 2010; Schutgens et al., 2014). Similarly, increasing vegetation complexity significantly reduced detection, consistent with other studies (Aschwanden et al., 2018; Morrison, 2002). Human detection probabilities also declined as the morning progressed, at faster survey speeds, and faster wind speeds. These findings suggest even mild environmental disturbances and observer fatigue may divert attention and reduce detection probability (Domínguez del Valle et al., 2020; Riding and Loss, 2018). While we did not find temperature to be an important predictor of human detection of carcasses, it is important to note that all surveys were conducted in temperatures ranging from 9 to 28 °C, and more extreme temperatures may impact performance. Overall, our findings reinforce the importance of visual cues in human detection and highlight the need to account for survey conditions and observer fatigue when designing carcass surveys to maximize detection.

Contrary to our predictions, detection of avian carcasses by the dog team was primarily influenced by the same factors that affected human detection—carcass size and ground substrate. Although previous studies report mixed effects of carcass size on detection for dogs (Reyes et al., 2016; Smallwood et al., 2020), our results showed a strong positive relationship: larger carcasses were more detectable, likely due to greater odor emission (Sloan et al., 2025), although visual detection by the dog may also be greater, particularly on bare ground. Detection probabilities by the dog team, however, remained high compared to human surveyors across carcass size classes (Fig. 6A). Previous studies have found variable relationships between ground substrate and detection of carcasses by dogs (Barrientos et al., 2018; Domínguez del Valle et al., 2020; Mathews et al., 2013), but we found detection significantly

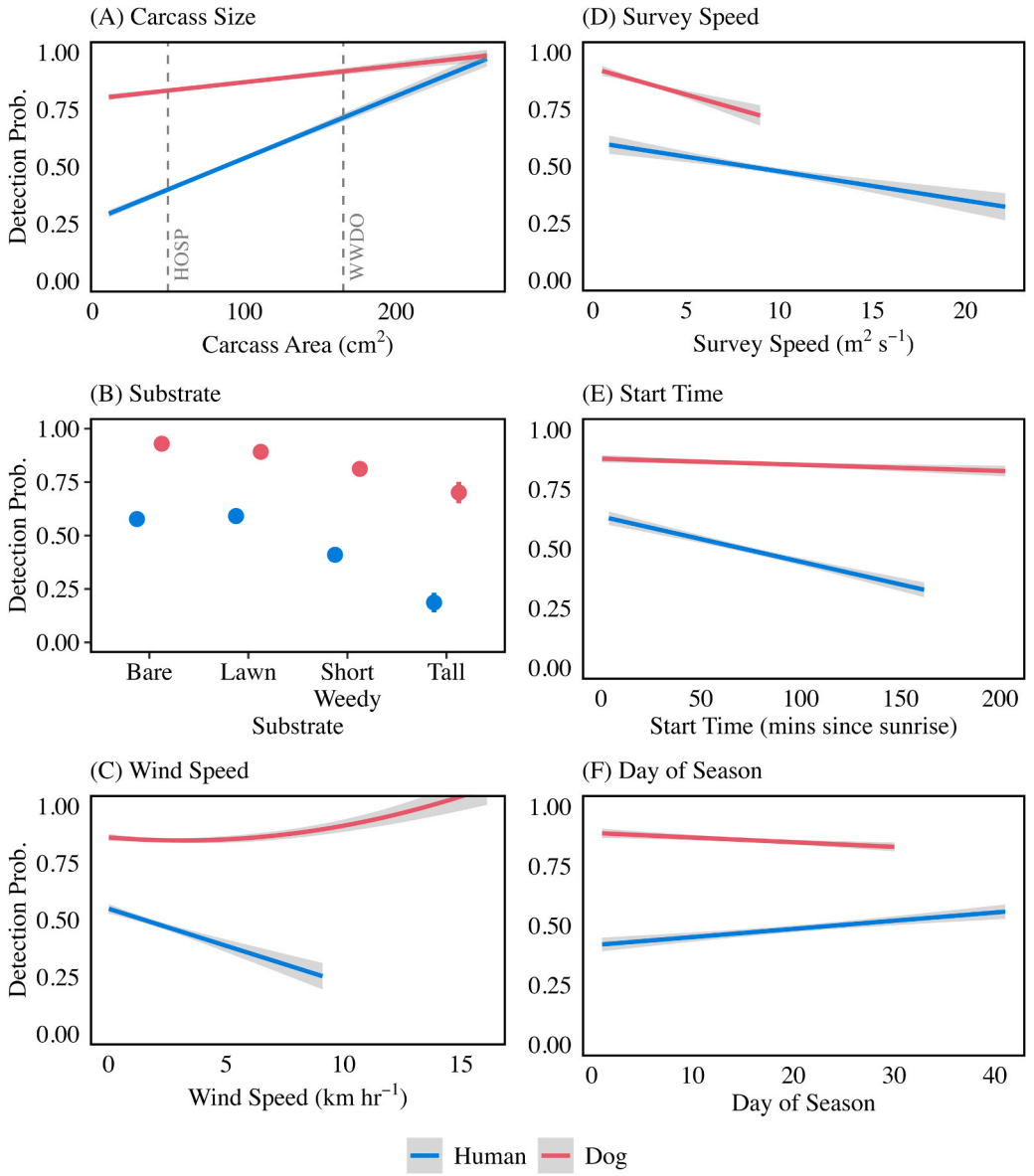


Fig. 6. Model output assessing factors influencing detection probability of avian carcasses by human surveyors and dog team in southern New Mexico, USA, fall 2022 and 2023. Predicted relationships between avian carcass detection probability by humans (blue) or dog (red) and (A) carcass size, with dashed lines indicating mean size of white-winged dove *Zenaida asiatica* (WWDO) and house sparrow *Passer domesticus* (HOSP) carcasses; (B) ground substrate; (C) wind speed; (D) survey speed; (E) survey start time; and (F) day of season. Gray shaded areas represent 85% confidence intervals.

decreased as vegetative complexity increased. Dense vegetation, especially the tall weeds and thorny shrubs present in some of our survey locations, may limit physical movement of dog surveyors and odor dispersal (MacKay et al., 2008; Mathews et al., 2013; Paula et al., 2011). Surprisingly, factors typically associated with scent dispersal, such as wind speed and temperature, showed little effect on detection by the dog. Although we observed a quadratic effect of wind on dog detection probability, this pattern appeared to be driven by a few surveys conducted under windier conditions (approximately 15 km hr^{-1}). Because surveys took place in early mornings when weather was generally mild, the range of wind speeds and temperatures was narrow and likely limited our ability to detect strong effects from these variables. Our use of fresh carcasses likely also limited scent availability. Furthermore, warm, dry conditions at our survey sites possibly reduced dog scenting ability by causing panting and drying of nasal tissues (MacKay et al., 2008; Reyes et al., 2016). As with human surveyors, detection by the dog team decreased as survey speed increased, an expected pattern, although not evaluated by other studies; this highlights the importance of a skilled handler who maintains an appropriate pace during surveys.

We relied on paid undergraduate students, who conducted surveys 1–5 times per week, in contrast to many carcass detection studies that employ full-time professionals. Although not statistically significant, our descriptive analysis suggested that detection

probabilities of our human surveyors improved over the 2022 season, while remaining consistent in 2023, suggesting enhanced training and mentorship may offset inexperience. Interestingly, only one of five returning surveyors showed a significant increase in detection, although the small sample size limited our ability to thoroughly investigate further. This is a potential avenue for future research as the literature presents conflicting conclusions about the role of surveyor experience in carcass detection; Rivera-Milán et al. (2004) found inexperienced and experienced teams performed equally well and Ponce et al. (2010) identified prior experience as an important predictor of detection. Individual observer variation (e.g., visual acuity, attention span) may also influence detection probabilities, though this was not directly tested in our study.

Management decisions in natural resources often involve trade-offs between competing objectives (e.g., cost and effort) in which the optimal choice may vary depending on local conditions or other uncertainties; design of carcass surveys is no exception (e.g., Barrientos et al., 2018; Smallwood, 2013). Based on our findings, we offer considerations for maximizing detection probability under different scenarios commonly faced by managers. While the dog team offered higher detection probabilities compared to humans across all conditions, their use is limited to temperatures under 28 °C for the safety of the dog (Slotta-Bachmayr et al., 2024). However, managers could implement modifications for human survey teams to maximize detection. Our results suggest that slow survey (walking) speeds in mild weather for a short duration are predicted to increase detection probability— mean human detection probability fell below 0.50 after approximately 75 min of surveying. Individual human detection varied widely; thus, managers could implement pre-survey detection trials to select only skilled surveyors. Additionally, the quality of training humans receive prior to surveying may also influence detection probability; detection probability of human surveyors started higher in the second year of our study when the human surveyor trainer (KCB) was more experienced and returning human surveyors could serve as additional mentors.

Individuals designing surveys may also want to consider the degree of uncertainty related to survey location and timing, as well as the ultimate goals of the survey when deciding among surveyor types. Effectiveness of both surveyor types could be influenced by carcass size as well as survey substrate (vegetation structure and terrain). For situations in which intended site(s) are known, such as regular surveys at fixed sites (e.g., wind farms), managers could decide if the improvement in detection probability that they could gain through dogs is sufficient, given the substrate and expected carcass species composition. Though our inference is limited to the substrate types we surveyed, several categories we included are broadly generalizable beyond our study area, such as pavement that is widespread in urban areas, and short vegetation cover that is reflective of manicured lawns in suburban areas or mowed grass near wind or solar installations. Similarly, the predictability of survey needs may also influence the choice between human and dog teams. Use of detection dogs often requires advanced planning, making them well-suited for regular monitoring, whereas episodic surveys (e.g., mass mortality events during migration or extreme weather) may be hindered by mobilization delays, increasing the risk of carcass loss to scavengers, and may be more well-suited for human surveyors. Lastly, the goal of the surveys may also influence the importance of maximizing detection probability, and the associated mortality estimate precision. Maximizing detection may be particularly important for studies seeking detailed species-level data or trends over time, collecting carcasses for testing (e.g., disease prevalence), detecting rare or newly invasive species (e.g., Fuller et al., 2024), or in contexts subject to high regulatory oversight. Recent advances in mortality estimation and reporting tools (e.g., estimators that explicitly account for imperfect detection and carcass persistence) can further support these efforts, but their performance ultimately depends on the quality of field data used to inform them (Dalthorp et al., 2018). In summary, while there is no “one size fits all” approach to carcass surveys, developing standardized core methods and reporting practices would greatly improve cross-study comparisons, although factors related to the location, timing, predictability, and the ultimate goal of surveys are important to consider.

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CRediT authorship contribution statement

Kelley C. Boland: Conceptualization, Methodology, Investigation, Writing – original draft, Formal analysis, Writing – review & editing, Visualization, Project administration. **Abigail J. Lawson:** Methodology, Formal analysis, Writing – review & editing, Visualization. **Dylan M. Osterhaus:** Writing – review & editing, Visualization. **Patricia L. Cutler:** Conceptualization, Writing – review & editing, Funding acquisition. **Gregory A. Davidson:** Investigation, Writing – review & editing. **Martha J. Desmond:** Conceptualization, Methodology, Writing – original draft, Formal analysis, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: GD and his dog team were contracted to perform carcass surveys, however GD was not involved in data analysis or the initial manuscript drafting. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.gecco.2026.e04148.

Data availability

Data will be made available on request.

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